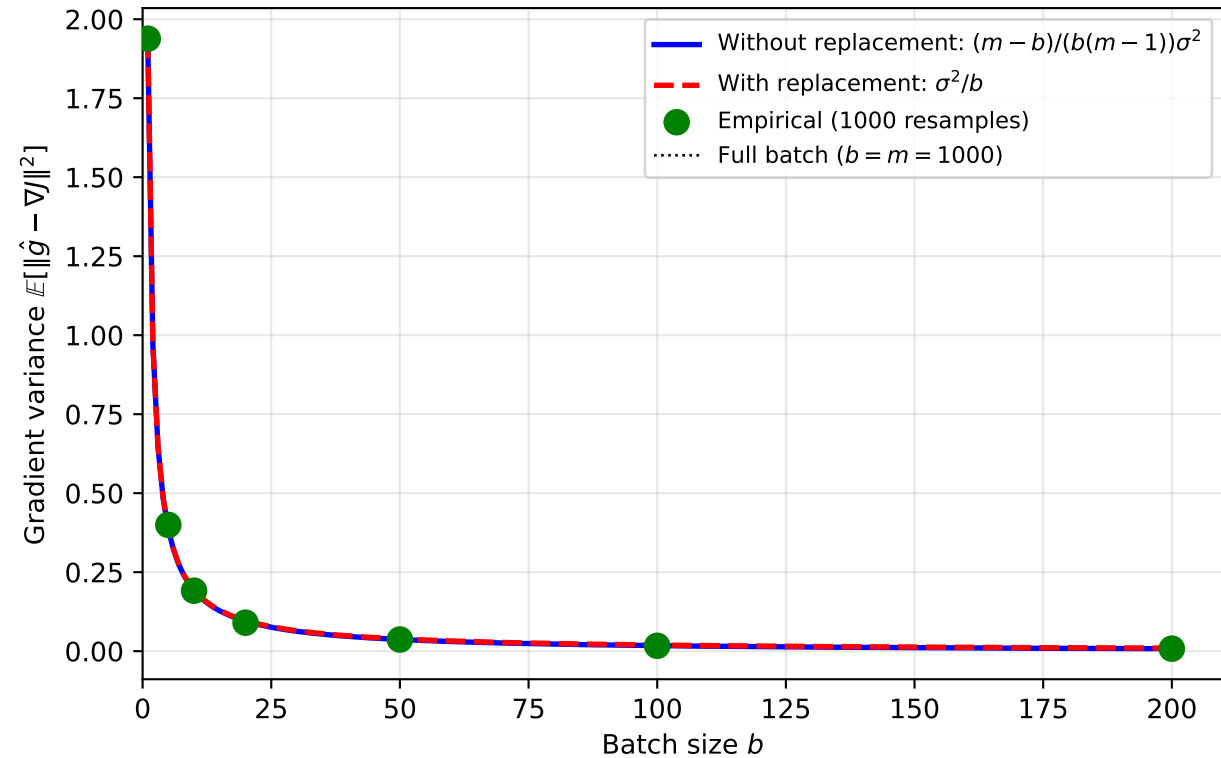


SGD Gradient Variance: Finite Population Correction $(m - b)/(b(m - 1))$

Mini-Batch SGD Gradient Variance vs Batch Size
(Benchmark A, $m = 1000$, $d = 2$)



Gradient Variance vs Batch Size (log--log)
W/R variance $\propto 1/b$; W/O/R variance $\rightarrow 0$ at $b = m$

